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Student Number

2018

Mathematics

HSC ASSESSMENT TASK 2

TUESDAY 20TH MARCH

General Instructions

- Reading time – 5 minutes
- Working time 3 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-16 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question
- Diagrams are not to scale

Total Marks – 100

Section I - Pages 3 - 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II - Pages 7 - 18

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Question	Marks
1 - 10	/10
11	/15
12	/15
13	/15
14	/15
15	/15
16	/15

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 30% of the Higher School Certificate Course Assessment

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Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 10 (Detach from paper)

- 1 What is 0.02095 rounded to 3 significant figures?
(A) 0.020
(B) 0.021
(C) 0.0209
(D) 0.0210

- 2 What is/are the solution(s) to the equation $x(x - 1) = x$?
(A) $x = 0$
(B) $x = 1$
(C) $x = 0$ and $x = 2$
(D) $x = 1$ and $x = 2$

- 3 Given $f(x) = x^6 + 3x^3 - 4x + 2$,
 $f''(x) =$
(A) $6x^5 + 3x^2 - 4$
(B) $6x^5 + 9x^2 - 4$
(C) $30x^4 + 6x$
(D) $30x^4 + 18x$

- 4 If $f'(x) = x^2 - 5x - 6$ then stationary points may occur when:
(A) $x = 1, -6$
(B) $x = -2, -3$
(C) $x = -1, 6$
(D) $x = -3, 2$

5 For what values of x is the curve $y = x^3 + 6x^2 - x + 4$ concave up?

(A) $x < -2$

(B) $x > 0$

(C) $x < 0$

(D) $x > -2$

6 Find $\int (2x + 1)^{\frac{1}{3}} dx$.

(A) $-\frac{2}{3}(2x + 1)^{-\frac{1}{3}} + C$

(B) $\frac{3}{4}(2x + 1)^{\frac{4}{3}} + C$

(C) $\frac{3}{8}(2x + 1)^{\frac{4}{3}} + C$

(D) $\frac{8}{3}(2x + 1)^{\frac{4}{3}} + C$

7 Evaluate $\int_0^9 (\sqrt{x} + 1)^2 dx$.

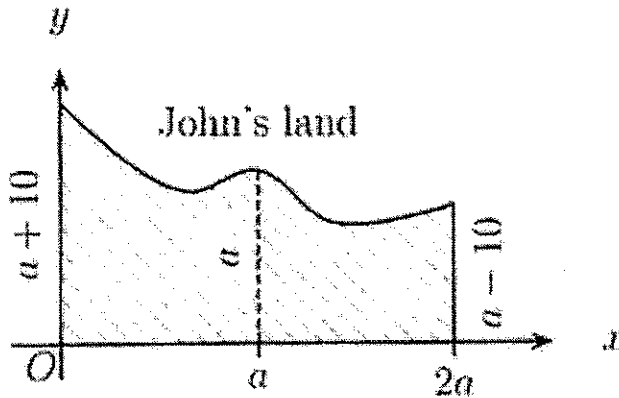
(A) $29\frac{1}{4}$

(B) $40\frac{1}{2}$

(C) $49\frac{1}{2}$

(D) $85\frac{1}{2}$

- 8 The shaded area shown in the diagram below represents John's land. Its dimensions are given in terms of a .
Given that the area of this land is $3\,200\text{ m}^2$, use Simpson's rule with 3 function values to find an estimate for the value of a .



- (A) $20\sqrt{2}\text{ m}$
 (B) $40\sqrt{2}\text{ m}$
 (C) 40 m
 (D) 1600 m
- 9 Differentiate $2 \tan 3x$.
- (A) $2 \sec^2 3x$
 (B) $6 \sec^2 3x$
 (C) $\frac{2}{3} \sec^2 3x$
 (D) $6 \sec 3x$
- 10 Evaluate $\int_0^{\frac{\pi}{8}} 4 \cos 4x \, dx$
- (A) 1
 (B) $\frac{1}{\sqrt{2}}$
 (C) -4
 (D) 4

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Section II

90marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11(15 marks) Use a SEPARATE writing booklet

- (a) Factorise completely $2c^2 - 50d^2$ 2
- (b) Rationalise the denominator of $\frac{6}{\sqrt{17} - \sqrt{11}}$ 2
- (c) Solve $|3x - 7| = 17$ 2
- (d) Simplify $\frac{5}{y+2} - \frac{2}{y}$ 2
- (e) The midpoint of the points A(3,-4) and B(x, y) is M(5,-1). Find the coordinates of B. 2
- (f) Solve the simultaneous equations 2

$$2x + y = 2$$

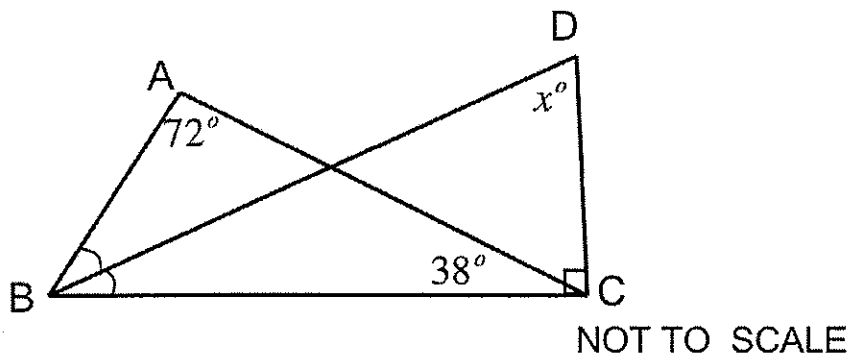
$$5x - y = 19$$

(g) In the diagram below (not drawn to scale), BD bisects $\angle ABC$,

$DC \perp BC$, $\angle BAC = 72^\circ$, $\angle ACB = 38^\circ$ and $\angle BDC = x^\circ$.

Copy the diagram into your writing booklet. Find the value of x , giving reasons for each step in your calculation.

3



End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Indicate and shade the region in which $4 < x^2 + y^2 \leq 9$

3

- (b) A function $f(x)$ is given by

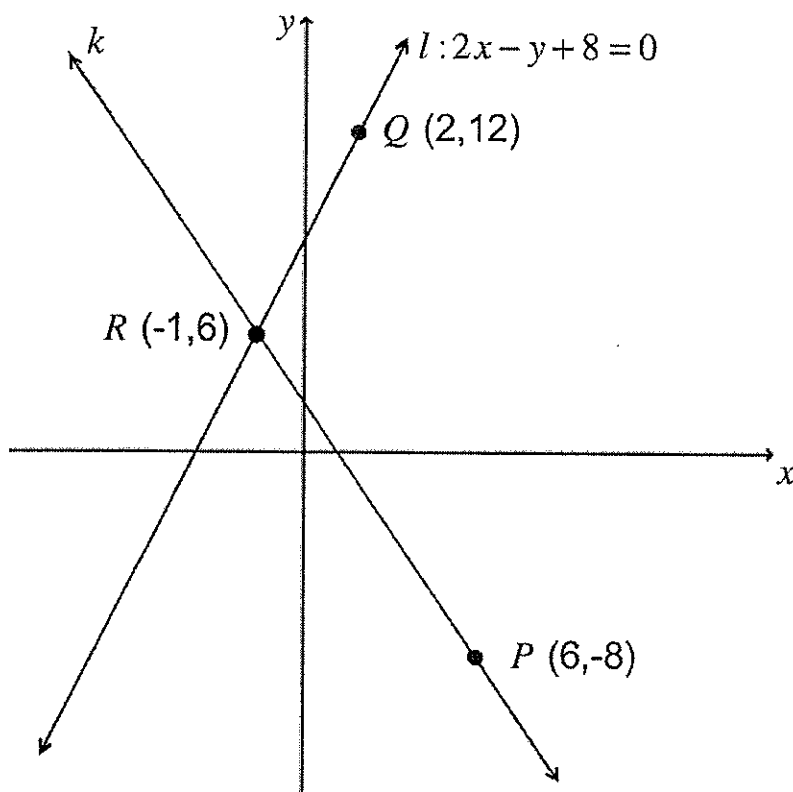
2

$$f(x) = \{x - 5, \text{ for } x \leq 3\}$$

$$\{x^2 - 16, \text{ for } x > 3\}$$

Find $f(3) - f(5)$

- (c)



The diagram above shows the line $l: 2x - y + 8 = 0$ and the point $Q(2, 12)$ on it. The line k has gradient -2 and passes through the point $P(6, -8)$. The lines k and l intersect at R .

$\uparrow (6, -8)$

- (i) Show that the equation of the line k is $2x + y - 4 = 0$.

1

- (ii) Show that the length of QR is $3\sqrt{5}$.

1

- (iii) Find the perpendicular distance from P to the line l . Leave your answer in surd form.

2

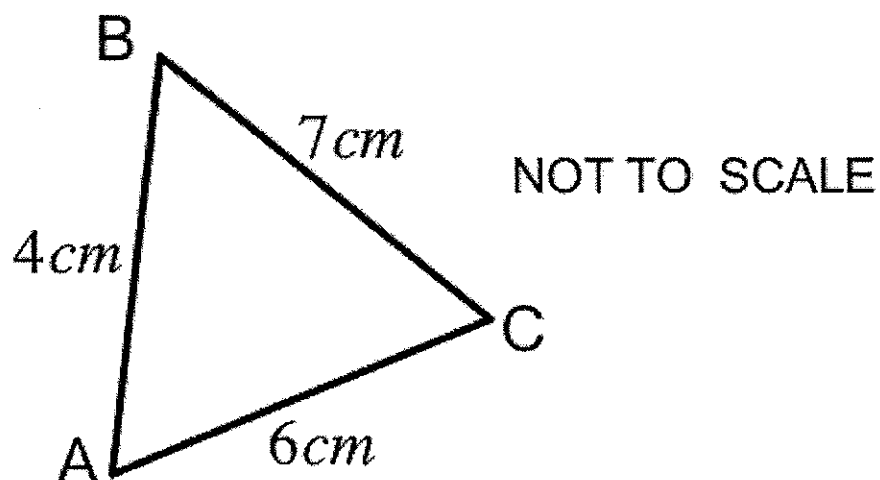
- (iv) Find the area of the triangle PQR .

1

Question 12 (continued)

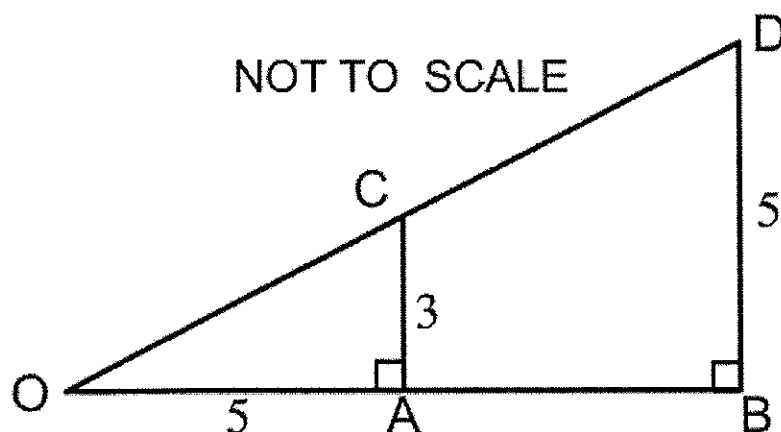
(d)

1



In the diagram above, ABC is a triangle in which $AB = 4\text{cm}$, $BC = 7\text{cm}$ and $CA = 6\text{cm}$. Use the cosine rule to show that $\cos C = \frac{23}{28}$.

(e)



In the diagram above, AC is parallel to BD, $OA = 5\text{cm}$, $DB = 5\text{cm}$ and $AC = 3\text{cm}$.

(i) Show that triangles OCA and ODB are similar.

2

(ii) Hence, find the length of AB, giving reasons.

2

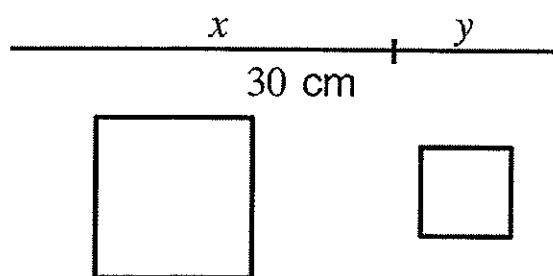
End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

a. Consider the curve $y = 3x^4 - 16x^3 + 24x^2$.

- i. Find the stationary turning points. 2
- ii. Hence determine the nature of the stationary points. 3
- iii. Sketch the curve, showing the intercepts with axes, stationary turning points and points of inflexion. 2

b. A 30 cm length of wire is cut into 2 pieces and each piece bent to form a square.



- i. Show that $y = 30 - x$ 1
 - ii. Show that the total area of the 2 squares is given by $A = \frac{x^2 - 30x + 450}{8}$ 2
- c. A truck travels 1 500 km at an hourly cost given by $s^2 + 9000$ cents where s is the average speed of the truck.
- i. Show that the cost for the trip is given by $C = 1500 \left(s + \frac{9000}{s} \right)$. 2
 - ii. Find the speed that minimises the cost of the trip. 3

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

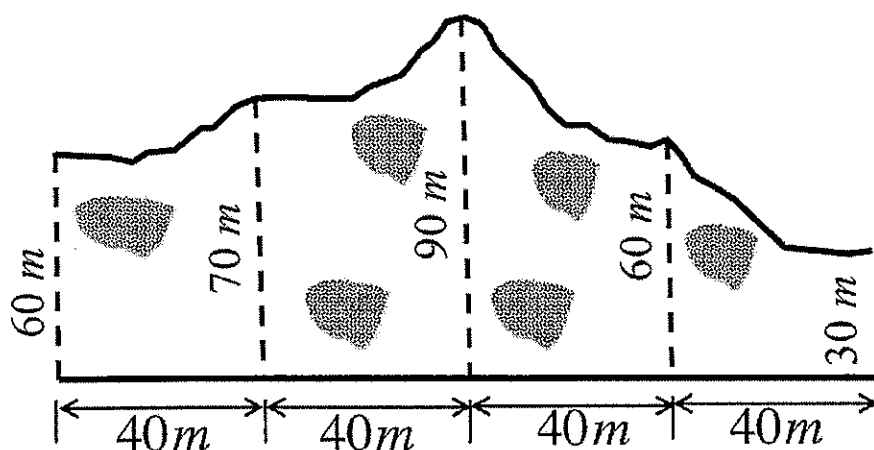
a. Find indefinite integral of $\int (x - 1)(x - 2) dx$.

2

b. Evaluate $\int_0^1 \frac{x^3 + 2x^2}{x} dx = \int_0^1 (x^2 + 2x) dx$.

2

c. The diagram shows the land that Jane bought near a lake.



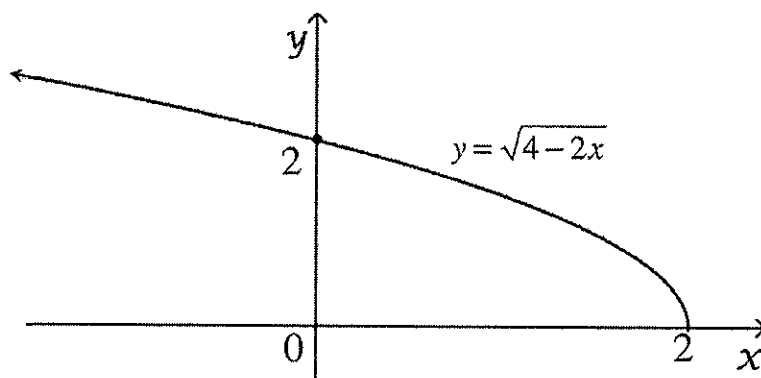
i. Use the trapezoidal rule with 5 function values to find an estimate for the area of this land.

2

ii. Is the area obtained in the previous part smaller or larger than the exact area? Give reasons for your answer.

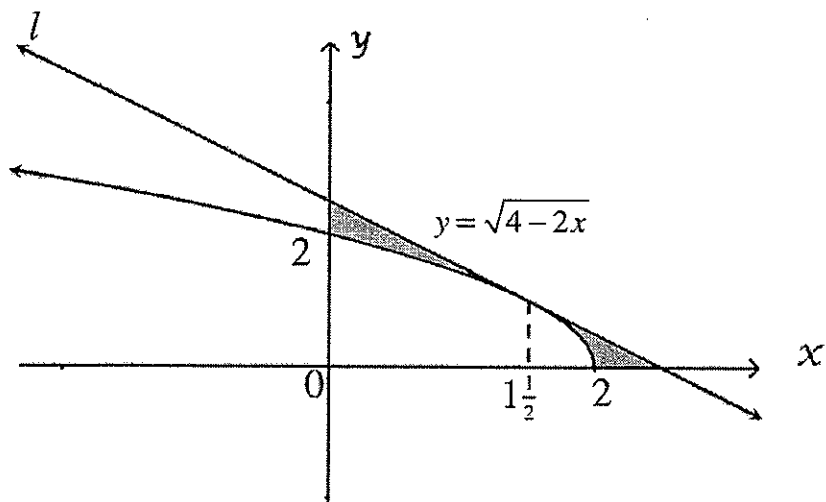
1

- d. The diagram below shows the graph of the curve $y = \sqrt{4 - 2x}$.



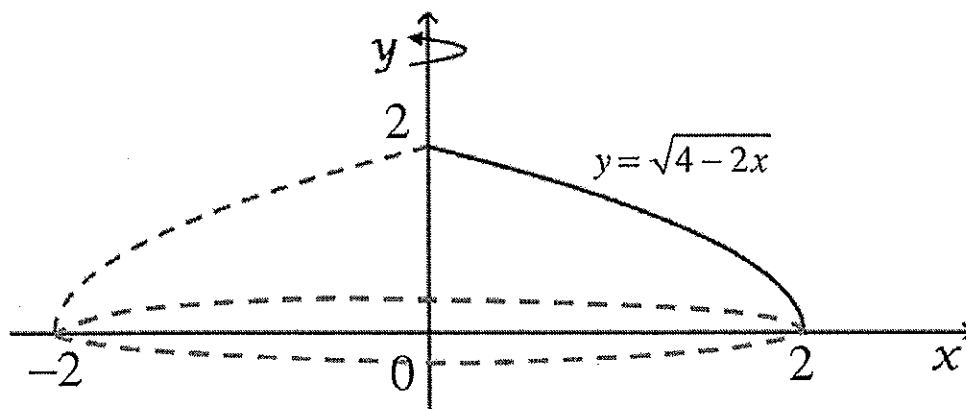
Question 14 Continued

- i. Show that the equation of the tangent l at the point on the curve where $x = 1\frac{1}{2}$ is $2x + 2y - 5 = 0$ 2
- ii. Show that the area bounded by the curve, the x axis and the y axis is $2\frac{2}{3}$ square units. 2
- iii. Using the result above, or otherwise, find the area of the region shaded between $y = \sqrt{4 - 2x}$, the tangent l , and the x and y axes as shown. 2



- iv. A spaceship is obtained by rotating part of the curve $y = \sqrt{4 - 2x}$ between the points $(2, 0)$ and $(0, 2)$ about the y axis as shown.

2



Find the volume of the spaceship.

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet

(a) Prove that $(\cot\theta + \operatorname{cosec}\theta)^2 = \frac{1+\cos\theta}{1-\cos\theta}$ 2

(b) Let α and β be the roots of $3x^2 - 8x + 3 = 0$.

(i) Find $\alpha\beta$ 1

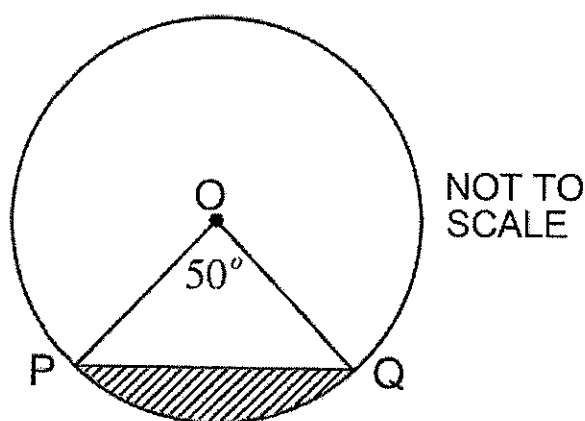
(ii) Hence find $\alpha + \frac{1}{\alpha}$ 1

(c) A parabola has equation $4y = x^2 - 4x + 8$

(i) Find the coordinates of its vertex. 2

(ii) Find the equation of its directrix. 1

(d)

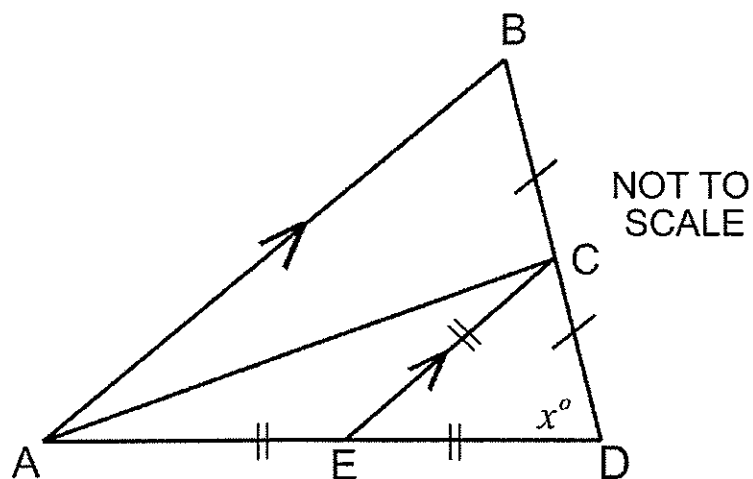


The diagram above, shows an arc PQ, of length 25π cm subtending an angle of $\frac{5\pi}{18}$ radians at O, the centre of the circle.

(i) Show that the radius of the circle is 90cm. 2

(ii) Calculate the shaded area in the diagram. (Give your answer to 1 decimal place) 2

- (e) In the diagram below, ABD is a triangle, the point C is the mid-point of BD and the point E is the mid-point of AD. Also, $AE = CE = DE$, $AB \parallel EC$ and $\angle EDC = x$



- | | | |
|------|--|----------|
| (i) | Show that $\angle ACB = 90^\circ$ | 3 |
| (ii) | Express $\angle BAD$ in terms of x . | 1 |

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet

(a) (i) Prove that $\frac{d}{d\theta}(\tan^3\theta) = 3\sec^4\theta - 3\sec^2\theta$ 2

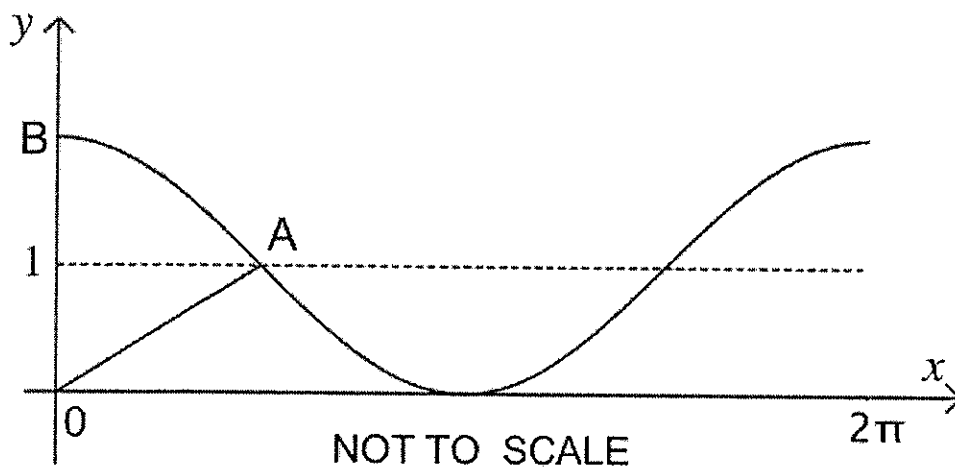
(ii) Hence evaluate $\int_0^{\frac{\pi}{4}} \sec^4\theta \, d\theta$ 2

(b) Given the function $y = -2\sin 3x$

(i) State the period of the function 1

(ii) Sketch the graph of $y = -2\sin 3x$ for $0 \leq x \leq \pi$. 2

(c)



The diagram shows the graph of $y = 1 + \cos x$, for $0 \leq x \leq 2\pi$

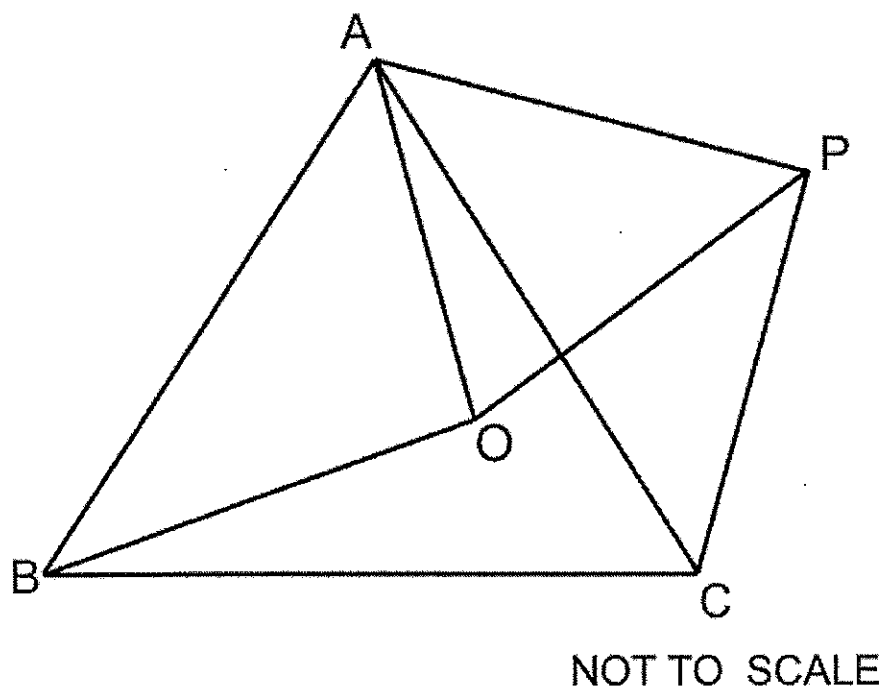
The graph crosses the line $y = 1$ at A, and the y-axis at B.

(i) Show that the y coordinate of A is $\frac{\pi}{2}$. 1

(ii) Show that the equation of OA is $y = \frac{2}{\pi}x$. 1

(iii) Hence find the area bounded by the arc AB, the y-axis, and the line OA. 2

(d)



In the figure above, triangles ACB and APO are equilateral.

- (i) Copy this diagram into your answer booklet and include all the given information.
- (ii) Explain why $\angle BAO = \angle PAC$ 2
- (iii) Prove that $\triangle AOB \equiv \triangle APC$ 2

End of Examination



Student Number

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 10. Detach this sheet.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

A B C D

A  B  C  D 

1.	A ☐	B ☐	C ☐	D ☐
2.	A ☐	B ☐	C ☐	D ☐
3.	A ☐	B ☐	C ☐	D ☐
4.	A ☐	B ☐	C ☐	D ☐
5.	A ☐	B ☐	C ☐	D ☐
6.	A ☐	B ☐	C ☐	D ☐
7.	A ☐	B ☐	C ☐	D ☐
8.	A ☐	B ☐	C ☐	D ☐
9.	A ☐	B ☐	C ☐	D ☐
10.	A ☐	B ☐	C ☐	D ☐

Sdn



Killara
HIGH SCHOOL

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Student Number

Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 10. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct

Start Here →

1. A ☐ B ☐ C ☐ D ☒
2. A ☐ B ☐ C ☒ D ☐
3. A ☐ B ☐ C ☐ D ☒
4. A ☐ B ☐ C ☒ D ☐
5. A ☐ B ☐ C ☐ D ☒
6. A ☐ B ☐ C ☒ D ☐
7. A ☐ B ☐ C ☐ D ☒
8. A ☐ B ☐ C ☒ D ☐
9. A ☐ B ☒ C ☐ D ☐
10. A ☒ B ☐ C ☐ D ☐

Multiple-Choice Solutions

1. 0.0210 (D)

2. $x(x-1) = x$

$$x^2 - x = x$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$x = 0$ and $x = 2$ (C)

3. $f(x) = x^6 + 3x^3 - 4x + 2$

$$f'(x) = 6x^5 + 9x^2 - 4$$

$$f''(x) = 30x^4 + 18x$$
 (D)

4. $f'(x) = x^2 - 5x - 6$

$$f'(x) = 0$$

$$x^2 - 5x - 6 = 0$$

$$(x+1)(x-6) = 0$$

$$x = -1, x = 6$$
 (C)

5. $y = x^3 + 6x^2 - x + 4$

$$y' = 3x^2 + 12x - 1$$

$$y'' = 6x + 12$$

The graph concave up when $y'' > 0$

$$6x + 12 > 0$$

$$6x > -12$$

$$x > -2$$
 (D)

6. $\int (2x+1)^{\frac{1}{3}} dx = \frac{(2x+1)^{\frac{4}{3}}}{\frac{4}{3} \times 2} + C$

$$= \frac{(2x+1)^{\frac{4}{3}}}{\frac{8}{3}} + C$$

$$= \frac{3}{8} (2x+1)^{\frac{4}{3}} + C$$
 (C)

7. $\int_0^9 (\sqrt{x} + 1)^2 dx = \int_0^9 (x + 2x^{\frac{1}{2}} + 1) dx$

$$= \left[\frac{x^2}{2} + \frac{4}{3} x^{\frac{3}{2}} + x \right]_0^9$$

$$= \left[\frac{81}{2} + \frac{4}{3} (9)^{\frac{3}{2}} + 9 \right] - 0$$

$$= 85\frac{1}{2}$$
 (D)

8.

x	0	a	2a
y	a+10	a	a-10
	y ₀	y ₁	y ₂

$$A = \frac{h}{3} (y_0 + 4 \sum y_{\text{odd}} + 2 \sum y_{\text{even}} + y_2)$$

$$3200 = \frac{a}{3} ((a+10) + 4(a) + (a-10))$$

$$3200 = \frac{a}{3} (6a)$$

$$2a^2 = 3200$$

$$a^2 = 1600$$

$$\therefore a = \pm 40$$
 (C)

9. Differentiate $2 \tan 3x$

$$\frac{d}{dx} (2 \tan 3x) = 1 \times 3 \sec^2 3x$$

$$= 6 \sec^2 3x$$
 (B)

10. $\int_0^{\frac{\pi}{8}} 4 \cos 4x dx = \left[\sin 4x \right]_0^{\frac{\pi}{8}}$

$$= \sin\left(\frac{4\pi}{8}\right) - \sin 0$$

$$= 1 - 0$$

$$= 1$$
 (A)

Q 11

$$\begin{aligned} (a) \quad 2c^2 - 50d^2 &= 2(c^2 - 25d^2) \\ &= 2[c^2 - (5d)^2] \\ &= 2(c-5d)(c+5d) \end{aligned}$$

1 mark for
partial factorisation

Another mark for
complete factorisation

$$(b) \quad \frac{6}{\sqrt{17} - \sqrt{11}} \times \frac{\sqrt{17} + \sqrt{11}}{\sqrt{17} + \sqrt{11}}$$

← 1 mark for showing
this step.

$$= \frac{6(\sqrt{17} + \sqrt{11})}{17 - 11}$$

$$= \sqrt{17} + \sqrt{11}$$

← 1 mark for correct answer.

$$(c) \quad |3x - 7| = 17$$

$$3x - 7 = \pm 17 \quad \leftarrow \text{1 mark}$$

$$3x = 7 \pm 17$$

$$x = \frac{7 \pm 17}{3}$$

$$= 8, -\frac{10}{3} \quad \leftarrow \text{1 mark}$$

$$(d) \quad \frac{5}{y+2} - \frac{2}{y} = \frac{5y - 2(y+2)}{y(y+2)}$$

← 1 mark for
working

$$= \frac{3y - 4}{y(y+2)}$$

or

$$\frac{3y - 4}{y^2 + 2y}$$

← 1 mark for
answer.

Q 11 (continued)

(e) $\frac{x+3}{2} = 5$ $\frac{y-4}{2} = -1$

$x = 7$ 1 mark

$y = 2$ 1 mark.

$\therefore M(7, 2)$

(f) $5x - y = 19$
 $2x + y = 2$

$7x = 21$

$6 + y = 2$

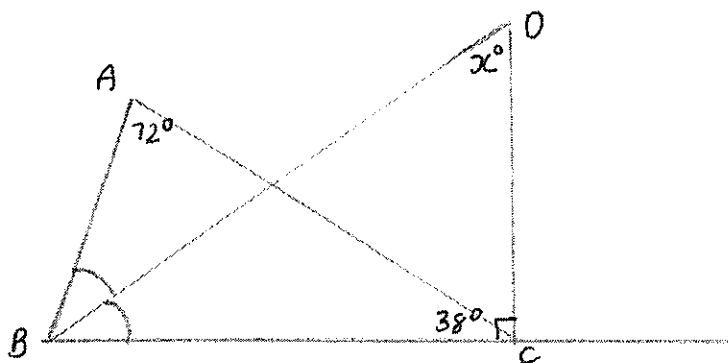
1 mark for working

$x = 3$

$y = -4$

$\therefore$ pt of intersection is $(3, -4)$ 1 for correct answer.

(g)



$\angle ABC + 38 + 72 = 180$ (Angle sum $\triangle ABC$) 1 mark

$\angle ABC = 70^\circ$

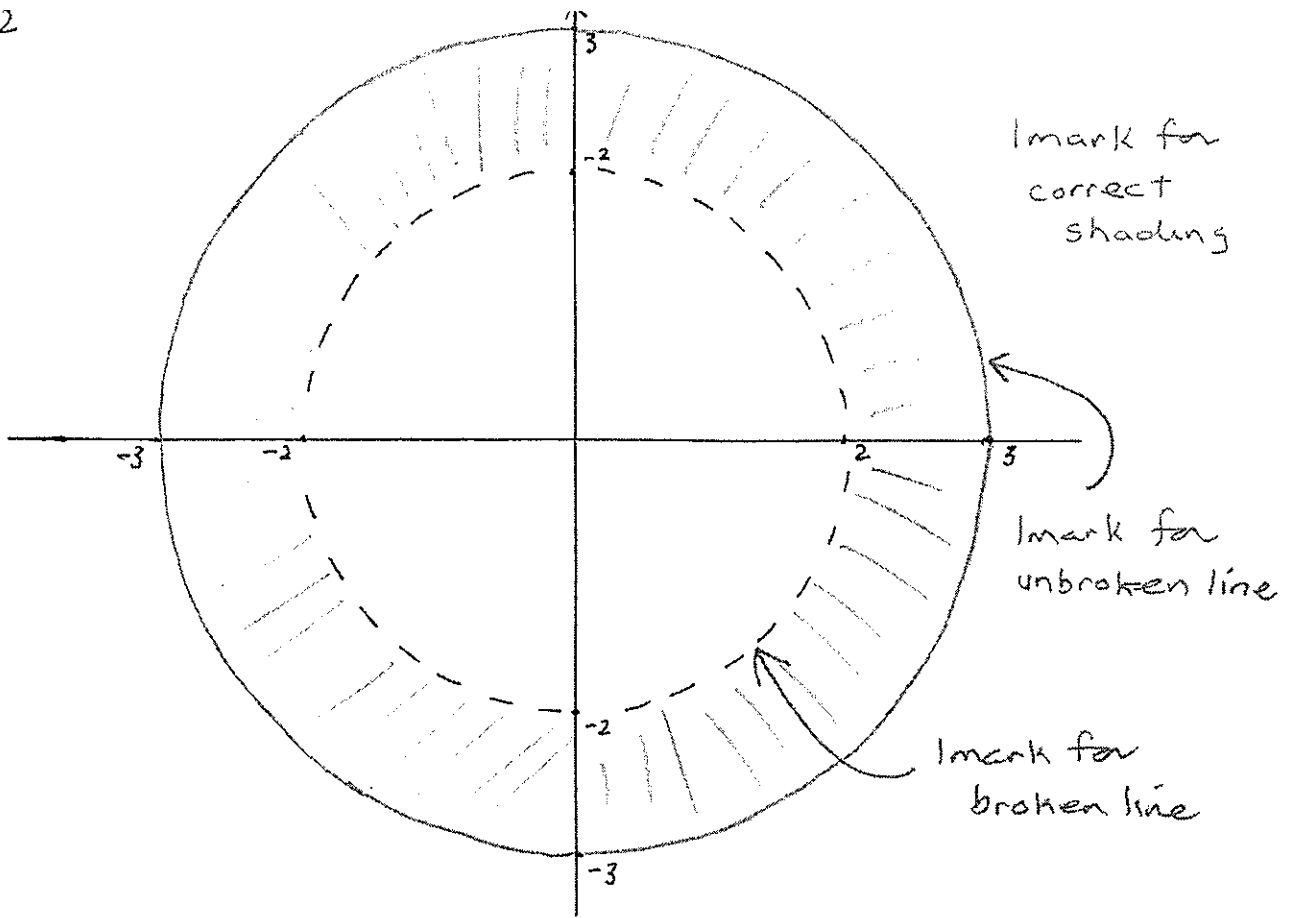
$\angle DBC = 35^\circ$ (DB bisects $\angle ABC$) 1 mark

$x + 90 + 35 = 180$ (Angle sum $\triangle BDC$) 1 mark

$x = 55^\circ$

Q12

a)



b)

$$f(3) - f(5)$$

$$= (3-5) - (5^2-16)$$

$$= -2 - 9$$

$$= -11$$

Mark for correct working

1 for correct answer

Q 12 (continued)

$$\begin{aligned} \text{(c) (i)} \quad y+8 &= -2(x-6) \\ y+8 &= -2x+12 \\ 2x+y-4 &= 0 \end{aligned}$$

Must show full working to receive mark.

$$\begin{aligned} \text{(ii)} \quad QR &= \sqrt{(2-(-1))^2 + (12-6)^2} \\ &= \sqrt{9+36} \\ &= \sqrt{45} \\ &= 3\sqrt{5} \end{aligned}$$

must show working

$$\begin{aligned} \text{(iii)} \quad \text{Distance } Pl &= \frac{|2 \times 6 + (-1) \times -8 + 8|}{\sqrt{4+1}} \\ &= \frac{12+8+8}{\sqrt{5}} \\ &= \frac{28}{\sqrt{5}} \end{aligned}$$

✓ 1 mark for working

✓ 1 mark for correct answer.

$$\text{(iv)} \quad \text{Area of } \triangle PQR = \frac{1}{2} \times 3\sqrt{5} \times \frac{28}{\sqrt{5}}$$

$$= 42 \text{ u}^2$$

✓ correct answer

$$\begin{aligned} \text{(d)} \quad \cos C &= \frac{a^2+b^2-c^2}{2ab} \\ &= \frac{7^2+6^2-4^2}{2 \times 7 \times 6} \\ &= \frac{49+36-16}{84} \\ &= \frac{23}{28} \end{aligned}$$

Must apply Cosine rule showing sufficient working.

(e)

$$\begin{array}{lcl}
 (i) & \angle OAC = \angle OBD = 90^\circ \text{ (given)} & \checkmark \\
 & \angle COA = \angle DOB \text{ (common angle)} & \checkmark
 \end{array}
 \left. \vphantom{\begin{array}{l} \angle OAC = \angle OBD = 90^\circ \text{ (given)} \\ \angle COA = \angle DOB \text{ (common angle)} \end{array}} \right\} \text{1 mark}$$

$$\therefore \triangle OCA \parallel \triangle ODB \text{ (equiangular)} \quad \text{1 mark}$$

$$(ii) \quad \frac{AC}{BD} = \frac{OA}{OB} \quad \text{(Sides of similar } \triangle's \text{ in same ratio)}$$

$$\therefore \frac{3}{5} = \frac{5}{5+AB}$$

↑
Award
1 mark

$$15 + 3AB = 25$$

$$3AB = 10$$

$$\therefore AB = \frac{10}{3} \text{ cm} \quad \leftarrow \text{Award 1 mark}$$

$$(3) y = 3x^4 - 16x^3 + 24x^2$$

$$(i) f(x) = 3x^4 - 16x^3 + 24x^2$$

$$\begin{aligned} f'(x) &= 12x^3 - 48x^2 + 48x \\ &= 12x(x^2 - 4x + 4) \\ &= 12x(x-2)^2 \end{aligned}$$

$$f'(x) = 0$$

$$12x(x-2)^2 = 0$$

$$x = 0, x = 2$$

$$f(0) = 0$$

$$\begin{aligned} f(2) &= 3(2)^4 - 16(2)^3 + 24(2)^2 \\ &= 48 - 128 + 96 \\ &= 16 \end{aligned}$$

$\therefore (0, 0)$ and $(2, 16)$ are the stationary points.

(ii)

$$\text{For } x = 0, y = 0$$

$$x = 2, y = 16$$

x	0	2
$\frac{dy}{dx}$	-	+
y	$\searrow (0, 0)$	$\nearrow (2, 16)$

$(2, 16)$ is a horizontal point of inflexion

$(0, 0)$ is minimum turning point

(iii)

Points of inflexion occur

$$\text{When } \frac{d^2y}{dx^2} = 0 = f''(x)$$

$$\begin{aligned} f''(x) &= 36x^2 - 96x + 48 \\ &= 12(3x^2 - 8x + 4) \\ &= 12(3x-2)(x-2) \\ &= 0 \end{aligned}$$

← 1 mark for differentiation.

← 1 mark for finding stationary points.

← 2 marks to determine the nature of the points

← 1 mark for H.P.O.I. and identify $(0, 0)$ is a minimum turning point.

ii) Continue

$$x = \frac{2}{3} \quad \text{or} \quad x = 2$$

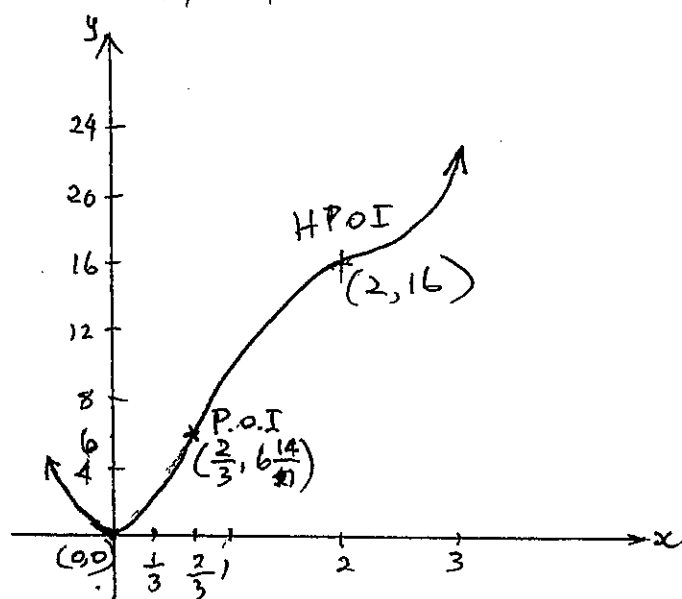
$$f\left(\frac{2}{3}\right) = 3\left(\frac{2}{3}\right)^4 - 16\left(\frac{2}{3}\right)^3 + 24\left(\frac{2}{3}\right)^2$$

$$= 6\frac{14}{27}$$

$$f(2) = 3(2)^4 - 16(2)^3 + 24(2)^2$$

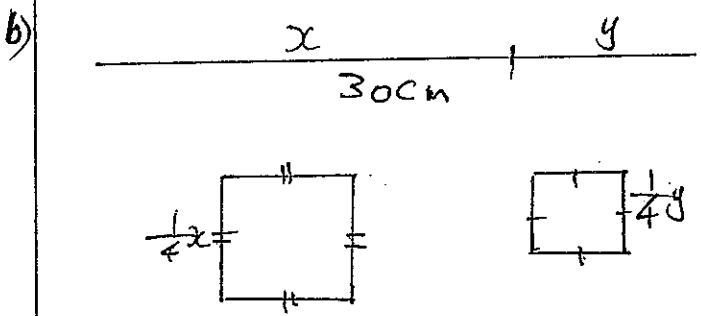
$$= 16$$

x	0	$\frac{2}{3}$	1	2	3
$\frac{d^2y}{dx^2}$	+	0	-	0	+
y	0	$6\frac{14}{27}$	16	16	0



← 1 mark (2 points must show full working to receive mark)

← 1 mark for correct graph.



i)

$$x + y = 30$$

$$y = 30 - x$$

← 1 mark for showing
 $y = 30 - x$ ✓

ii) Total area of the two squares

$$= \left(\frac{1}{4}y \times \frac{1}{4}y\right) + \left(\frac{1}{4}x \times \frac{1}{4}x\right)$$

$$= \left(\frac{1}{4}y\right)^2 + \left(\frac{1}{4}x\right)^2$$

$$= \frac{1}{16}y^2 + \frac{1}{16}x^2 \quad \text{Since } y = 30 - x$$

← 1 mark for correct substitution.

$$\therefore A = \frac{(30-x)^2}{16} + \frac{1}{16}x^2$$

$$= \frac{900 - 60x + x^2 + x^2}{16}$$

$$= \frac{900 - 60x + 2x^2}{16}$$

$$= \frac{450 - 30x + x^2}{8}$$

$$= \frac{x^2 - 30x + 450}{8}$$

← 1 mark for finding the total area.

$$\text{Hourly Cost} = s^2 + 9000 \text{ cents}$$

$$\text{Distance } d = 1500 \text{ km}$$

$$\text{Show } C = 1500 \left(s + \frac{9000}{s} \right)$$

$$t = \frac{\text{Distance}}{\text{speed}} = \frac{1500 \text{ km}}{s}$$

Total cost for the trip = hourly cost $\times$ time t

$$C = (s^2 + 9000 \text{ Cents}) \frac{1500 \text{ km}}{s}$$

$$= 1500 \left(\frac{s^2}{s} + \frac{9000}{s} \right)$$

$$= 1500 \left(s + \frac{9000}{s} \right)$$

← 1 mark for correct substitution

← 1 mark for working

Find the speed that minimizes the cost of the trip.

$$\text{Since } C = 1500 \left(s + \frac{9000}{s} \right)$$

$$\therefore C = 1500s + \frac{13500000}{s}$$

$$\frac{dC}{ds} = 1500 - \frac{13500000}{s^2}$$

$$\frac{dC}{ds} = 0$$

$$1500s^2 = 13500000$$

$$s^2 = 9000$$

$$s = 94.87$$

$$s \doteq 95 \text{ km/h}$$

← 1 mark for differentiation.

← 1 mark for correct working.

← 1 mark for correct answer

14a) $\int (x-1)(x-2) dx = \int (x^2 - 3x + 2) dx \leftarrow 1 \text{ mark for expanding}$
 $= \frac{x^3}{3} - \frac{3x^2}{2} + 2x + C \leftarrow 1 \text{ mark for integrating.}$

b) $\int_0^1 \frac{x^3 + 2x^2}{x} dx = \int_0^1 (x^2 + 2x) dx$
 $= \left[\frac{x^3}{3} + x^2 \right]_0^1 \leftarrow 1 \text{ mark for integrating}$
 $= \left[\frac{1}{3} + 1 \right] - 0$
 $= \frac{4}{3}$
 $= 1 \frac{1}{3} \leftarrow 1 \text{ mark for correct answer.}$

c)

x	0	40	80	120	160
y	60	70	90	60	30
	y_0	y_1	y_2	y_3	y_4

Trapezoidal rule

$A \div \frac{h}{2} (y_0 + 2(y_1 + y_2 + y_3) + y_4) \leftarrow 1 \text{ mark for substituting 5 function values.}$
 $\div \frac{40}{2} (60 + 2(70 + 90 + 60) + 30)$
 $\div 10600 \text{ m}^2 \leftarrow 1 \text{ mark for correct answer.}$

(ii) It can be seen that the trapezoidal Rule gives an area greater than the actual area. (as the sum of the areas of the 4 trapeziums is greater than the actual area.) $\leftarrow 1 \text{ mark for giving the reasons.}$

d) $y = \sqrt{4-2x} = (4-2x)^{\frac{1}{2}}$

(i) $m = \frac{dy}{dx} = \frac{1}{2} (4-2x)^{-\frac{1}{2}} \times (-2) = -\frac{1}{\sqrt{4-2x}}$
 at $x = \frac{1}{2} = \frac{3}{2}$

$m_{\text{tangent}} = -\frac{1}{\sqrt{4-2(\frac{3}{2})}} = -1$

When $x = \frac{3}{2}$ $y = (4-2(\frac{3}{2}))^{\frac{1}{2}} = 1$
 $\therefore x = \frac{3}{2}, y = 1$

$\leftarrow 1 \text{ mark for finding the co-ordinate } x = \frac{3}{2}, y = 1.$

ii) Continue
using the point-gradient formula

$$(y-1) = -1(x - \frac{3}{2})$$

$$y-1 = -x + \frac{3}{2}$$

$$y = -x + \frac{5}{2}$$

$$2y = -2x + 5$$

$$\therefore 2x + 2y - 5 = 0$$

1 mark for using the point-gradient formula to show the equation of the tangent l.

i) To find the area $\int_0^2 (4-2x)^{\frac{1}{2}} dx$

$$= \left[-\frac{1}{3}(4-2x)^{\frac{3}{2}} \right]_0^2$$

$$= -\frac{1}{3}(0 - 4^{\frac{3}{2}})$$

$$= -\frac{1}{3}(-8)$$

$$= \frac{8}{3}$$

$$= 2\frac{2}{3} \text{ units}^2$$

1 mark for integrating

1 mark correct working.

iii) The equation of the tangent is
 $2x + 2y - 5 = 0$

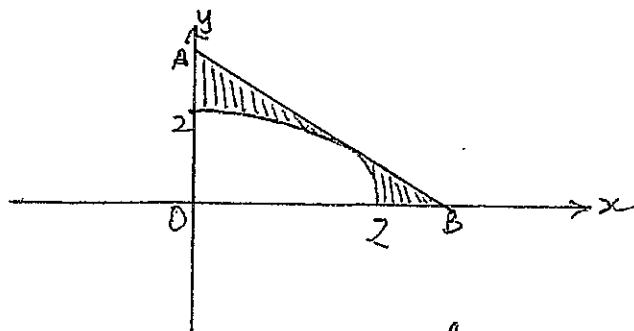
To find the coordinates of A and B.

Let $x=0$ to give $y = \frac{5}{2}$

$y=0$ to give $x = \frac{5}{2}$

$$\therefore \text{Area of } \triangle OAB = \frac{1}{2} \times \frac{5}{2} \times \frac{5}{2} = \frac{25}{8}$$

1 mark for finding the area of $\triangle OAB$.



The area under curve found in Part (ii) is
 $A = \frac{8}{3}$

$\therefore$ The required area is

$$A_{\triangle OAB} - A = \frac{25}{8} - \frac{8}{3} = \frac{11}{24} \text{ units}^2$$

1 mark to find the area of the shaded region.

d) Find the volume of the spaceship.
10) We need to write x^2 in terms of y . Changing the subject to x^2

$$y = \sqrt{4 - 2x}$$

$$y^2 = 4 - 2x$$

$$2x = 4 - y^2$$

$$x = 2 - \frac{y^2}{2}$$

$$x^2 = \left(2 - \frac{y^2}{2}\right)^2$$

$$= 4 - 2y^2 + \frac{1}{4}y^4$$

The volume of the spaceship is

$$V = \pi \int_0^2 x^2 dy$$

$$= \pi \int_0^2 \left(4 - 2y^2 + \frac{1}{4}y^4\right) dy$$

$$= \pi \left[4y - \frac{2}{3}y^3 + \frac{1}{20}y^5\right]_0^2 \leftarrow \text{1 mark for integrating.}$$

$$= \pi \left[8 - \frac{16}{3} + \frac{32}{20}\right]$$

$$= \frac{64\pi}{15} \text{ units}^3$$

$\leftarrow$ 1 mark correct answer.

Q 15

SOLUTIONS

(a)

$$\frac{\text{LHS}}{(\cot \theta + \operatorname{cosec} \theta)^2} = \frac{\text{RHS}}{\frac{1 + \cos \theta}{1 - \cos \theta}}$$

$$\frac{\text{LHS}}{= \left(\frac{\cos \theta}{\sin \theta} + \frac{1}{\sin \theta} \right)^2}$$

1 mark awarded

$$= \left(\frac{1 + \cos \theta}{\sin \theta} \right)^2$$

$$= \frac{(1 + \cos \theta)(1 + \cos \theta)}{\sin^2 \theta}$$

$$= \frac{(1 + \cos \theta)(1 + \cos \theta)}{1 - \cos^2 \theta}$$

$$= \frac{\cancel{(1 + \cos \theta)}(1 + \cos \theta)}{\cancel{(1 + \cos \theta)}(1 - \cos \theta)}$$

1 mark awarded

$$= \frac{1 + \cos \theta}{1 - \cos \theta}$$

$$\therefore \text{LHS} = \text{RHS}$$

$$(b) (i) \alpha \beta = \frac{3}{3}$$

$$= 1$$

1 mark

$$(ii) \alpha \beta = 1$$

$$\beta = \frac{1}{\alpha}$$

$$\therefore \alpha + \frac{1}{\alpha} = \frac{8}{3}$$

1 mark

Q15 (continued) Solutions

(c)

$$(i) \quad x^2 - 4x + 8 = 4y$$

$$x^2 - 4x = 4y - 8$$

$$x^2 - 4x + (-2)^2 = 4y - 8 + 4$$

1 mark awarded

$$(x-2)^2 = 4(y-1)$$

$\therefore$ Vertex is $(2, 1)$

1 mark for
correct answer

(ii) Since the focal length = 1,

the equation of directrix is $y = 0$ 1 mark

(d)

$$(i) \quad l = r\theta$$

$$25\pi = r \times \frac{50\pi}{180}$$

← 1 mark

$$r = \frac{25 \times 180}{50}$$

← 1 mark

$$= 90 \text{ cm}$$

$$(ii) \quad \text{Shaded area} = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times 90^2 \left(\frac{5\pi}{18} - \sin\left(\frac{5\pi}{18}\right) \right) \checkmark \quad 1 \text{ mark}$$

$$= 431.8 \text{ cm}^2 \quad (1 \text{ decimal pl})$$

↑
1 mark for
correct rounding

Q 15 continued Solutions

e)(i) In $\triangle ECD$, $\angle ECD = x$ (angles opposite equal sides in isosceles $\triangle$.)
 $\therefore \angle CEA = 2x^\circ$ (exterior angle of $\triangle CED$)

1 mark

In $\triangle AEC$, $\angle ECA = \angle EAC$ (angles opposite equal sides in isosceles $\triangle$)
 $= \frac{180-2x}{2}$ (Angle sum of $\triangle AEC$)
 $= 90-x$

1 mark

Now $\angle ACB = 180 - (\angle AEC + \angle ECD)$ (straight angle)
 $= 180 - (90-x+x)$
 $= 90^\circ$

1 mark

(ii) $\angle BAD = \angle CED$ (Corresponding angles, $AB \parallel EC$)
and $\angle CED = 180-2x$ (Angle sum of $\triangle CED$) ✓
 $\therefore \angle BAD = 180-2x$

1 mark

Question 16 Solutions

$$\begin{aligned}
 a) (i) \frac{d}{d\theta} (\tan^3 \theta) &= 3 \tan^2 \theta \cdot \sec^2 \theta && \text{1 mark} \\
 &= 3(\sec^2 \theta - 1) \sec^2 \theta \\
 &= 3 \sec^4 \theta - 3 \sec^2 \theta && \text{1 mark}
 \end{aligned}$$

$$(ii) \int_0^{\frac{\pi}{4}} \sec^4 \theta \, d\theta$$

$$= \frac{1}{3} \int_0^{\frac{\pi}{4}} (3 \sec^4 \theta - 3 \sec^2 \theta) + 3 \sec^2 \theta \, d\theta \quad \text{1 mark}$$

$$= \frac{1}{3} \left[\tan^3 \theta + 3 \tan \theta \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{3} \left[\left(\tan^3 \frac{\pi}{4} + 3 \tan \frac{\pi}{4} \right) - 0 \right]$$

$$= \frac{4}{3}$$

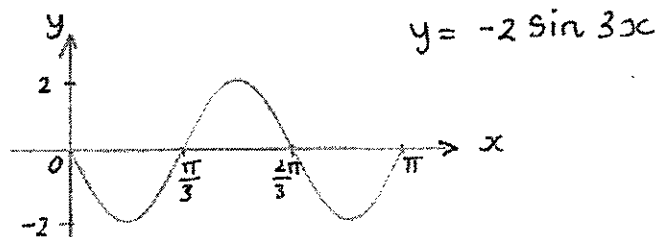
1 mark

b) (i) The period x

$$\begin{aligned}
 3x &= 2\pi \\
 x &= \frac{2\pi}{3}
 \end{aligned}$$

1 mark

(ii)



1 mark for correct curve

1 mark for correct labelling

Question 16 (continued)

(c) (i) When $y=1$,

$$1 + \cos x = 1$$

$$\cos x = 0$$

$$\therefore x = \frac{\pi}{2}$$

Show working
for 1 mark

(ii) Gradient $OA = \frac{1}{\frac{\pi}{2}}$

$$= \frac{2}{\pi}$$

Show working
for 1 mark.

$$\therefore OA : y = \frac{2}{\pi}x$$

(iii) $A = \int_0^{\frac{\pi}{2}} (1 + \cos x - \frac{2x}{\pi}) dx$

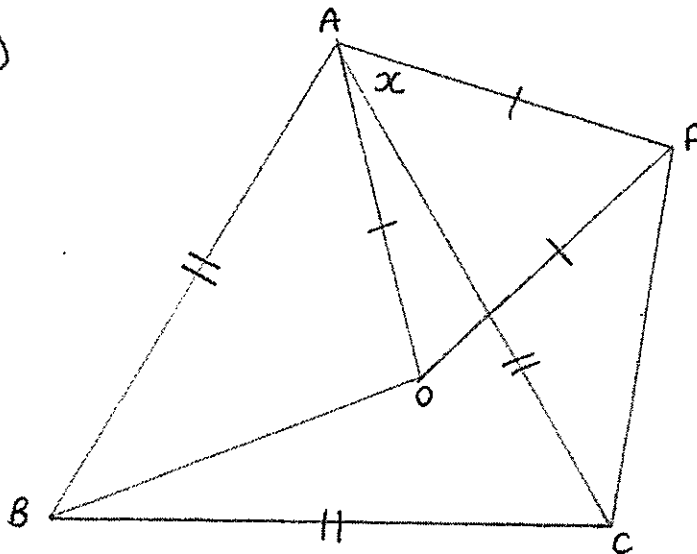
$$= \left[x + \sin x - \frac{1}{\pi}x^2 \right]_0^{\frac{\pi}{2}} \quad \checkmark \quad \text{1 mark.}$$

$$= \left[\frac{\pi}{2} + 1 - \frac{1}{\pi} \times \frac{\pi^2}{4} \right] - [0 + 0 - 0]$$

$$= \left(\frac{\pi}{4} + 1 \right) \text{ unit}^2 \quad \checkmark \quad \text{1 mark}$$

Question 16 (continued) Solutions

(d) (i)



ii) Let $\angle PAC = x$

$\therefore \angle CAO = 60 - x$ ($\triangle APO$ is equilateral)

$\angle BAC = 60$ ($\triangle ABC$ is equilateral)

$\angle BAO = 60 - (60 - x)$

$= x$

$\therefore \angle PAC = \angle BAO$

1 mark

1 mark

(iii) In $\triangle AOB$ & $\triangle APC$:

$AB = AC$ (sides of equilateral $\triangle ABC$)

$AO = AP$ (sides of equilateral $\triangle AOP$)

$\angle BAO = \angle CAP$ (shown above in (ii))

$\therefore \triangle AOB \equiv \triangle APC$ (SAS)

1 mark

1 mark

deduct 1 mark if
no diagram given.